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Deployment of an Indigenous Solution for Seamless Integration of Compact Portable Multi-Phase Flow Meter (MPFM) for Testing Shared Well Strings

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Abstract

Dual-string wells with a shared oil flowline present unique challenges in production testing, primarily due to sharing of flow lines and strings, inaccurate allocation, and operational constraints. Traditional methods such as test separators and production logging tools (PLTs) require well shutdown and production deferment. This paper presents the successful deployment of compact portable multiphase flow meters (MPFMs) without production deferment and resulting in 7% increase in sustainable oil production rate equivalent of 4500BOPD, in the one of the largest onshore fields.

The Central Degassing Station at one largest onshore fields processes crude from over 650 wells, contributing up to 30% of onshore production. Due to increased drilling, approximately 15% of the wells share common flow lines, with up to four wells sometimes connected to a single line. A pilot project was initiated using a compact MPFM in one such shared oil string without shutting down wells, in contrast to traditional methods involving test separators. The data was validated against traditional test separator measurements. The designs of all the sharing wells were studied, and customized mono block valve assemblies were added for the seamless hookup of compact MPFM to test the shared string without causing any production deferment.

The results demonstrate that portable MPFMs provide accurate and repeatable measurements while eliminating the need for test separators or well shut ins. Compared to PLTs, MPFMs enable continuous monitoring without intervention, significantly reducing downtime and operational risks. Additionally, compact portable MPFM deployment improved data acquisition frequency, supporting real-time decision-making for production optimization and well management. The findings underscore compact MPFMs' potential to enhance well testing efficiency, avoiding production deferment and sustainability while reducing environmental impact by minimizing flaring and emissions.

Deployment of compact MPFM were tested for 6 months and following benefits were realized.

- Increase in 7 % sustainable production equivalent to 4500 BOPD of oil.
- Well availability enhancement of shared oil strings (typically 2-4 strings are shared)

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- Equivalent annual saving of 82MM$ for production gain.
- Better reflection of accuracy as testing is done nearer to the flowing condition since sharing wells/ strings are not closed.
- GHG emission reduction of 6307 metric ton of Co2.

The implementation of an innovative technique for the seamless installation of compact Multiphase Flow Meters (MPFM) on dual strings with comingled flow lines represents a novel approach to minimizing production interruptions and enhancing well availability, compared to conventional well testing techniques. This methodology can be applied to other oil and gas operations that face challenges associated with shared well flow lines.

Introduction

The hydrocarbon gathering and processing infrastructure is composed of Remote Degassing Stations (RDSs) and a Central Degassing Station (CDS). Surface installations at RDS sites facilitate the oil and gas gathering and further transfer to centralized facilities via dedicated transfer lines. Multiphase Pumps (MPPs), installed at each gathering station, enable the transportation of unseparated production streams by elevating line pressure as required. The oil gathering network employs Multi-Port Selector Manifolds (MSMs) to direct flow from multiple wells to designated MPP stations, MPP units boost discharge pressure, ensuring transportation capability. Some of the RDS oil gather in the production header and sent to CDS via transfer lines. MPP is usded where wells don't have enough pressure.

Production fluids from three remote RDSs are directed through piggable pipelines to the CDS for further separation and stabilization. Typical RDS specifications include:

- Gathering manifolds for primary collection
- Testing skids for performance monitoring
- Transfer manifolds for redirected flows
- Chemical dosing systems for process optimization
- Pig launch/receive facilities for pipeline maintenance
- Flare/Ignition assemblies for controlled hydrocarbon disposal
- Instrument air supply systems for pneumatic actuation
- Fire & Gas detection and suppression modules
- Relief and drain networks for safe system depressurization

Operational protocol dictates that RDSs are manned during daylight hours and night operations are managed remotely via a Distributed Control System (DCS) integrated with Supervisory Control and Data Acquisition (SCADA) interfaces, providing continuous real-time process control in CDS.

Well completions encompass single, dual, and artificial lift categories. Reservoir drive mechanisms leverage water and gas injection schemes to sustain production rates. All fieldwork conforms to Well Integrity and Operational Standards aligned with statutory requirements for safety, environmental protection, ethical conduct, and fiscal responsibility.

The CDS receives and conditions hydrocarbon streams from over multiples wells. Inbound emulsions containing significant water cut and associated gases are processed in five parallel Gas Oil Separation Plant (GOSP) trains designed for sequential phase disengagement and crude stabilization. Each GOSP train accommodates feedstock with up to 30% volumetric water content.

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Process architecture for each train incorporates Dual-Stage Separation: an initial High Pressure (HP) separator facilitates gross gas-liquid fractionation, followed by an Intermediate Pressure (IP) three-phase vessel for discrete crude, water, and gas segregation. Trains 1–3 incorporate integrated two-stage electrostatic desalting and dehydration modules, employing chemical injection and high-voltage fields to achieve product specification. Electrostatic treaters enhance aqueous phase coalescence, expediting dewatering.

Trains 4 and 5 utilize surge vessels post-IP separation, allowing gravity-driven settling prior to transfer to heated, electrostatically augmented dehydrators. This modular approach supports flexibility in response to dynamic inlet characteristics, scheduled maintenance, and reliability objectives.

Comprehensive instrumentation—including level controls, pressure relief valves, and corrosion probes—ensures operational integrity. Pipelines convey treated crude, separated water, and recovered gases for storage, reinjection, or further processing, supporting maximized recovery, compliance with export quality standards, and minimization of flaring and effluent emissions.

Produced water management involves gravity-based free water removal in primary/secondary separators and demulsifier-assisted separation of emulsified phases via chemical and/or electrostatic methods.

Treated crude transitions through degassing boots (three in service), where lean gas stripping regulates Reid Vapor Pressure (RVP) and strips volatile fractions. Each degassing boot operates as an independent gas/liquid separator, capable of managing throughputs equivalent to two full GOSP trains.

Processed oil proceeds to Flow Suction Tanks (FSTs) for final sediment/water separation and buffer storage before being routed through a header system to five booster pumps. These escalate delivery pressure to three Main Oil Line (MOL) pumps for onward dispatch via parallel spur lines into the main export grid.

Associated gas at various pressure stages (HP/IP/surge/degassing/tank vapours) is consolidated and routed through stratified gas headers, with high-pressure streams supplied directly to the Gas Lift Compression Station (GLCS).

Produced water from all process stages is collected in four parallel disposal tanks (two new, two legacy), equipped with skimming apparatus for residual oil removal and transferred for deep well disposal or reinjection. Redundancy allows individual tank isolation for maintenance while sustaining continuous operation.

Reservoir pressure maintenance utilizes a gas injection plant comprising both gas turbine- and electrically-driven compressors, configurable based on grid and process requirements, with the turbine unit providing standby operability for enhanced reliability.

Field Background and Challenges

The Central Degassing Station at one of largest onshore hydrocarbon production facilities is a critical component in the collection and primary processing of crude oil, receiving inflow from over hundreds of individual wells distributed across several production clusters. The station currently accounts for approximately 30% of total onshore output, thereby playing a pivotal role in sustaining overall field deliverability and meeting corporate production quotas.

As field development activities have intensified, the operational paradigm has shifted towards increased wellbore density and shared infrastructure to optimise cost and surface facility footprint. Correspondingly, around 15% of producing wells are now configured to utilize common flow lines, with up to four discrete wells occasionally tied into a single trunk line. This configuration introduces significant complexity with respect to both reservoir management and surface operations.

Operationally, when a production test is required for any given string, current protocols necessitate that other dual strings—alongside any other wells sharing a common flowline—must be temporarily isolated and shut-in. This practice results in intermittent deferment of production corresponding to the number of affected strings, typically resulting in a loss ranging from 1,000 to 3,000 barrels per day (bbl/day), considering

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an average productivity index of 1,000 bbl/day per string. These deferred volumes directly impact Well Availability and Well Testing Key Performance Indicators (KPIs), which are essential performance metrics within company integrated asset management framework

This onshore field is mandated to align with well testing guidelines, which stipulate stringent requirements for periodic well testing and continuous verification of production allocations, as governed by Law #8 (specifically articles 20 & 29) and IRM (Integrity, Reliability, and Maintenance) production testing standards. These regulations require that each individual well undergoes valid, documented production tests at prescribed intervals. As such, the Well Testing KPI is not only a tactical operational metric but also a strategic compliance indicator, routinely scrutinized at the highest organizational levels to ensure robust tracking of individual well strings, major reservoir units, and aggregate field production potential.

The intersection of regulatory compliance, operational efficiency, and maximization of hydrocarbon recovery presents ongoing challenges, particularly under current constraints where testing of dual string completions and wells with shared surface infrastructure necessitates partial or complete shut-ins, adversely affecting daily production objectives. Additionally, the legacy measurement systems in place lack the granularity and isolation needed to obtain accurate data from individual wells connected via shared flowlines, which can lead to ambiguities in allocation and uncertainty in reservoir performance analysis.

To remedy these operational inefficiencies and support company broader sustainability and digital transformation objectives, the deployment of compact portable multiphase flow meters (MPFM) at the wellhead has been identified as a transformative solution. These devices enable real-time, in-situ measurement of phase-specific production rates—oil, gas, and water—at the source, thereby facilitating compliant well testing without necessitating operational shutdowns of adjacent strings or flowlines. The application of MPFM technology addresses critical gaps in both production measurement fidelity and test frequency, promotes optimal asset utilisation, and ensures company remains agile in fulfilling both regulatory mandates and internal performance targets. The integration of MPFM units represents a best-practice approach to modern production operations, positioning the field for enhanced data-driven decision-making, improved hydrocarbon allocation accuracy, and sustained long-term productivity.

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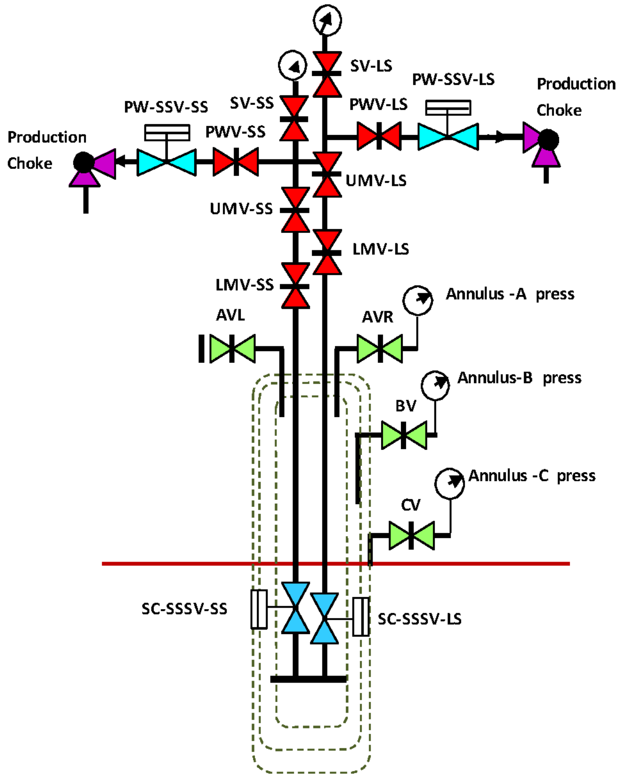

Legend:		
1	SV-SS	Short String Manual Swab Valve
2	PWV-SS	Short String Manual Production Wing Valve
3	PW- SSV-SS	Short String Actuator Surface Safety Production Wing Valve
4	UMV-SS	Short String Manual Upper Master Valve
5	LMV-SS	Short String Manual Lower Master Valve
6	SC-SSSV-SS	Short String Actuator Surface-Controlled Subsurface Safety Valve
7	SV-LS	Long String Manual Swab Valve
8	PWV-LS	Long String Manual Production Wing Valve
9	PW-SSV-LS	Long String Actuator Surface Safety Production Wing Valve-
10	UMV-LS	Long String Manual Upper Master Valve
11	LMV-LS	Long String Manual Lower Master Valve
12	SC-SSSV-LS	Long String Actuator Surface-Controlled Subsurface Safety Valve
13	AVR	Annulus A Manual Gate Valve Right
14	AVL	Annulus A Manual Gate Valve Left
15	BV	Annulus B Manual Gate Valve
16	CV	Annulus C Manual Gate Valve

Dual Completion Oil Well

Dual completion wells have the following configuration:

- Two tubing are used to extract oil from separate reservoir zones.
- Two production strings convey oil to the gathering manifold.
- The Xmas tree includes eight main gate-type valves (four per production string) and a wellhead cap.

Each tubing contains a Sub-Surface Safety Valve (SSSV), while each production string incorporates a Surface Safety Valve (SSV). The flow lines from each production string (designated as short/long strings) are merged into a single flow line, with check and isolation valves installed on each flow line. Choke valves —typically of the positive type—are installed on every production string for regulating flow.

The primary eight valves installed on the Xmas tree comprise:

- Two lower master valves (one per string) positioned at the lower section of the Xmas tree, serving as primary isolation valves.
- Two upper master valves (one per string) located above the lower master valves.

Wellhead Control Panel

The wellhead control panel (WHCP) manages the surface and sub-surface safety valves by supplying pressurized hydraulic oil for opening operations. Hydraulic pressure is generated by an electric motor-pump assembly (24 V DC motor) to operate these valves. Two hydraulic pressure ranges are configured within the WHCP. The control panel is powered by solar energy and housed in a passive cool shelter.

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Hydraulic System

The hydraulic system, managed from the WHCP, supports valve actuation. It includes an electric-driven pump and a hydraulic main reservoir. The pump provides necessary operating pressure, while the reservoir stores circulating hydraulic oil. All valve return lines connect to the reservoir inside the panel. Pressure regulators maintain the required pressure within actuation lines.

Solar System/Passive Shelter

The power supply for the panel is 24 VDC, -10% to +15%, with negative earth. This supply is delivered by an external solar power system through a non-redundant feeder for hydraulic pumps and the control panel. Solar power at the wellhead station supplies the wellhead control panel, PLC, SCADA panels, and limited lighting. An insulated enclosure serves as a passive shelter for housing PLC and SCADA equipment. A thermo-siphon based water cooling system regulates the shelter's internal temperature, enabling autonomous wellhead operation independent of utility infrastructure.

Control Description

Process control relies on the operation of surface and sub-surface safety valves. Signals from high and low-pressure transmitters connect to the PLC, which manages the SSV. High and low fusible plugs, hydraulically activated by the WHCP, depressurise their respective hydraulic loops during fire events, triggering closure of the SSV and SSSV to shut down the wellhead. Additional safety features include Emergency Shutdown switches, with a remote ESD point installed at least 30 metres from the wellhead—typically near a suitable location or at the well fence gate.

Deployment methodology

A pilot project was implemented to evaluate the deployment of a compact multiphase flow meter (MPFM) within a shared oil production string, specifically designed to facilitate continuous well testing without necessitating the shutdown of wells—a significant shift from conventional methodologies that rely on test separators for individual well assessment. This approach is particularly advantageous in environments where minimizing production deferment is critical to maintaining field productivity and meeting operational targets.

The MPFM's performance metrics were rigorously validated through comparative analysis against reference data obtained from traditional test separator measurements. This validation process encompassed an extensive review of the operating conditions, including pressure, temperature, flow rate, and fluid composition, ensuring the integrity and reliability of the MPFM readings under varying field scenarios.

To accommodate the integration of the compact MPFM into the shared string configuration, a comprehensive engineering review of all well completions in the shared system was performed. This entailed a detailed examination of tubing and casing designs, flow regimes, and valve schematics. Subsequently, monoblock valve assemblies were procured and installed at strategic locations along the wellhead and manifold infrastructure. These assemblies facilitated seamless connection and disconnection of the compact MPFM, thereby enabling flexible and efficient setup for periodic or ad hoc well testing campaigns, all while obviating the need for production interruption or deferred output.

Operationally, the portable compact MPFM was positioned at the inlet line immediately downstream of the chokes valve, optimizing measurement accuracy by capturing the multiphase flow prior to any significant phase separation or fluid conditioning. The meter's outlet was connected downstream of the loop valve, integrating the testing apparatus into the existing flow path with minimal disruption to established operations. This configuration enabled real-time acquisition of key production parameters—such as oil, water, and gas flow rates—for each contributing well in the shared string.

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Figure 1—Dual string well with compact MPFM connection

Figure 2—Field deployment of compact MPFM

This methodology stands in contrast to legacy practices wherein the shared string must be isolated and closed off to facilitate routing to Remote degassing station (RDS) or Centralized Degassing Station(CDS) test separators. In systems featuring commingled flow lines, such closures typically require cessation of production from multiple wells, with the potential to significantly compromise aggregate field output and hinder the achievement of daily production targets.

The successful deployment of this compact MPFM technology in shared production strings represents a substantial advancement in well testing strategies. By enabling accurate, on-the-fly well diagnostics without impeding overall production, operators can optimize reservoir management, enhance surveillance capabilities, and make more informed decisions regarding allocation, artificial lift optimization, and interventions. This innovation not only reduces operational downtime but also provides a scalable

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model for future implementations across assets characterized by complex well configurations and shared infrastructure.

Techno economical evaluation was carried out for both portable test separator and compact MPFM as shown in the below table. Compact MPFM was more feasible than the portable test separator.

Parameter	Portable Test Separator	Compact MPFM
Rig up & down	2 days	Few hours
Tests / month	10 to 12 tests	20 tests
Cost / month	100,000 USD	70 to 80,000 USD
Calibration	Results for verification	Results for verification Calibration inputs
Implementation	Current practice	Pilot Completed
Impact on Production	7% decrease in FSOPR	7% increase in FSOPR

The deployment of compact MPFM was more feasible and economically since it increased the FSOPR to 7%.

The subject received formal endorsement from Technical Committee Engineering (TC Eng) and was further supported by the Development Team (Dev), signifying alignment across key technical and operational stakeholders. The across field deployment of this scale up pilot is scheduled for execution in 2025, with preparatory activities and resource allocation currently underway to ensure readiness.

A field evaluation involving a mobile well test unit equipped with a Multiphase Flow Meter (MPFM) supplied by Haimo was conducted on July 7th at site. This trial was pivotal in mitigating an anticipated production deferment of 6,000 barrels of oil per day (BOPD) from sharing well. The results from this intervention are presently undergoing a comprehensive review by the Development Team to assess effectiveness and delineate lessons learned for future deployments.

The contractor's detailed budget estimate for procuring a dedicated, truck-mounted MPFM unit with all necessary accessories is about to complete this year, Concurrently, initiative registration under the centralized Technology Platform has been finalized, establishing a formal governance structure and ensuring compliance with internal innovation management processes.

Additionally, an across-field contract has been initiated to enable integrated flow line testing capabilities spanning multiple asset locations. Field deployment of these resources is actively progressing, with monitoring frameworks in place to gather performance data, validate equipment reliability, and inform recommendations for broader implementation. This systematic approach supports both operational excellence and maximization of hydrocarbon recovery through real-time multiphase measurement and reduced unplanned deferments.

Conclusion

The utilization of compact multiphase flow meters (MPFMs) installed on dual production strings with comingled flow lines facilitates uninterrupted, real-time measurement of hydrocarbon streams without necessitating operational shutdowns. This advanced configuration significantly enhances both operational efficiency and overall well availability by enabling continuous surveillance and optimization of production parameters. The deployment of MPFMs resulted in a substantial increase in oil output—specifically, a 7% uplift equating to approximately 4,500 barrels of oil per day (BOPD). Furthermore, this improvement in production capability translated into considerable cost savings, amounting to $82 million annually.

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Collectively, these advancements underscore the value of integrating MPFM systems within modern field developments, driving both economic and environmental gains while supporting robust operational integrity.

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Glossary

CDS : Central degassing station
BWPD : Barrels of water per day
GOSP : Gas oil separation plant
HP : High pressure
LP : Low pressure
IP : Intermediate pressure
MOL : Main oil line
FST : Flow suction tank
RVP : Reid vapor pressure
PT : Pressure transmitter
DWT : Disposal water tank
FWT : Fresh water tank

References

CDS operating manual, Well head and control system

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